Experiments in Physics: PHYS4350_01 Spring 2022
Boston College, Department of Physics
Syllabus

Scheduled class meeting time:
Tuesday and Thursday: 1:30 – 4:00 PM, Room = 130 and 250 Higgins Hall

Required Texts:
There is no textbook is required to be purchased for this course. Journal articles and other materials are required for this course and will be made available to students by the Instructor on Dropbox, email or in the Boston College library.

Instructor:
Fr. Cyril P. Opeil, S.J., Ph.D., Associate Professor of Physics
Office: St. Mary's Hall, Superior's Office, 617-552-8200 or 617-552-8171.
Office: Higgins 330J, 617-552-8450; Laboratory: Higgins 130, 617-552-3589

Office Hours:
Due to the COVID19 pandemic all office hours will be via Zoom. Please, email the Instructor at opeil@bc.edu to set up a Zoom meeting for Tuesday 4:00-5:30 PM or Thursday 4:00-5:30 PM. Offices hours are only offered during these times.

To contact Fr. Opeil, S.J. outside of class use email: opeil@bc.edu or you can call my secretary, Mrs. Kelly Cunniffe to leave a message for me, at 617-552-8200.

Graduate Assistant:
No graduate student will be assigned to this class.

Grading Exercises, Quizzes, Presentations and Laboratory Reports:
The instructor will grade all assignments for this class.

Laboratory Reports 10 points each
Kaleidagraph Exercise 5 points
Library Exercise 5 points
Archimedes Exercise part 1, 5 points
Archimedes Exercise part 2, 10 points
Cp gamma Debye Exercise 10 points
Reading quizzes 20 points
Student presentation 10 points

Pre-requisites: Senior/Junior Physics majors are eligible to take this class.

Course Description: PHYS4350_01 is a three-credit course designed for senior/junior Physics majors. The course consists of a series of lectures, exercises, reading assignments and laboratory exercises that focus on sample preparation, measurement

and analysis techniques used in physics research. The experiments are designed to enable hands-on exploration of laboratory techniques. The course will be a hybrid of Zoom lecture classes and in-class laboratory experiments.

Course Objectives: This course introduces students into experimental techniques of sample preparation and measurements associated with condensed-matter physics research. The techniques explored in this course are used by graduate students engaged in physics research with the intent of publication in refereed journals. Students will gain critical skills in performing laboratory techniques associated with sample making, sample preparation and temperature dependent measurements.

Course Goals:
1) Learn about the various measurement temperature and field dependent techniques used in condensed-matter physics research.
2) Experience, hands-on sample making and preparation techniques.
3) Perform temperature-dependent measurements on experimental samples.
4) Gather and analyze experimental data utilizing computers and test equipment.
5) A series of experimental research articles will be assigned for reading and general knowledge will be tested by a quiz. The research articles are not directly related to the assigned laboratory but are used to show the breath and detail of experimental techniques.
6) Two brief (10-12 minute) presentations will be conducted by a student on the assigned experimental technique, i.e. the Nernst Effect.

Reading Assignments/Quizzes: Scientific periodical literature is one of the major products of research and how people communicate in academics. In effort to expand the horizons of undergraduate students, scientific articles will be read throughout this course. The topics of these articles do not have a direct bearing or coordination on the experiments we will perform in this course. This is intentional. For students going on to graduate school in Physics, the experience of reading and comprehending articles will become part of your life. For students not going on to graduate school, these articles will give you an introduction to techniques beyond the time and equipment limitations of this course. A brief multiple-choice question quiz will be given on each reading according to the Reading Assignments schedule each Tuesday. Copies of assigned articles are available (only to students enrolled in this class, at no cost, for educational purposes to avoid copyright infringement). Articles will arrive either by Dropbox or email.

Exercises/Homework:
Several exercises and homework assignments will be due throughout the semester. These exercises assist the student in understanding a variety of experimental techniques.

Library Exercise
Kaleidagraph/Origin Exercise
C_p analysis Exercise
Archimedes Exercise Parts 1 & 2 (Part 1 is assigned at a particular time. Part 2 can be scheduled at any time. Ask the instructor for the apparatus.)

Student Presentations on Experimental Techniques:
One 12-15 minute Powerpoint (oral) presentation will be given by each student. The presentation will focus on a particular experimental technique. Topics will be assigned by the instructor.

Experimental Laboratory Procedures:
1. Using an ULVAC-ZEM to measure Seebeck and resistivity on Constantan and Nickel. [ZSR]
2. Speed of Sound in Metals at 295 K and Young's Modulus, multi-frequency probe. [SSMF]
3. Strain gage calibration and measurement. [SGCM]
4. Speed of Sound in materials at 295 K using a single frequency, through transmission. [SSTT]
5. Capacitance and Dielectric Measurements & Measuring Q in an tank circuit. [CQTT]

Laboratory Reports will be due one week after finishing the experiments in the laboratory.

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Safety: The laboratory exercises essential to in this course require that you perform them while observing appropriate safety rules, regulations and requirements. Laboratories can be dangerous places for people who do not observe safety rules, regulations and requirements. *The instructor will not tolerate students who do not observe appropriate rules, regulations and requirements. There will be no tolerance for inappropriate behavior on the part of students, because the inability to observe safety rules can result in serious permanent physical injury. If a student is asked to leave the laboratory during a laboratory experiment, for improper behavior, they will lose all points associated with that exercise.* Goggles and gloves will be required when working with materials or doing experimental work. No goggles or gloves are required to use computers in the laboratory. No one should work alone while in the laboratory. Commonsense and experience are the best defense against accidents, if you have little or no laboratory experience ask an experienced person before you perform laboratory tasks. Report any dangerous situations to the instructor immediately. Almost all laboratory accidents are preventable, the majority of accidents happen when people do not observe safety rules or do not apply their commonsense. If you have a future in experimental Physics, safety concerns will be part of your future. (Theorists have little to worry about beyond spilling their cappuccino on their keyboard.) Accidents happen, but liability issues for universities, companies and government agencies call for eliminating many problems.

Attendance, Participation and Academic Honesty: Continuity and active participation are crucial to the success of this course. You are responsible for all information from each lecture and laboratory session whether you attend or not. Absences due to (excused absence) sports activities are to be discussed with the instructor prior to the planned absence.

This course requires people to work individually on each experimental set up. Each experimental report will be written by that individual.

Please review the Boston College Standards of Academic Integrity that are published in the university catalog and on the web at:

http://www.bc.edu/offices/stserv/academic/resources/policy/#integrity.

It is *your responsibility* to ensure that all work you submit is in accordance with university policies. If you have any questions, please consult with the instructor. Violations will be reported to the Deans' Office and reviewed by the College's Committee on Academic Integrity. Sanctions for violating this policy include the possibility of failing this course.

Class participation/Attendance: Students are expected to attend class and perform assigned laboratory exercises. Do not be afraid to ask questions. Despite the pandemic you need to attend class in person. No laboratory experiments can be done on line.

Dropbox: Assigned readings from journals, laboratory procedures, exercises, periodic announcements, the syllabus and scheduling adjustments can be found on a Dropbox link sent by the instructor to the students via email. Students will be required to sign up for a free Dropbox account. Go to "https://www.dropbox.com" on the www and just to go "sign up for free".

Laboratory Hours:
Laboratory space and equipment are available for students during the posted laboratory hours. Use your time effectively. Please, do not request personal special laboratory hours. Safety rules mandate that student laboratories must be supervised, requiring the presence of the instructor or teaching assistant. The instructor and laboratory assistants are available only during normal laboratory hours.

Disabilities: If you have a disability, please discuss the situation with the instructor. A laboratory is not a traditional classroom so accommodations to the laboratory environment may invite creative solutions. Laboratory courses require certain skills and accommodations can be made to assist students when possible and necessary.

Eating and Drinking: *Absolutely* no food or beverages are to be brought into or consumed in the lecture room or laboratory. No water bottles! Not consuming food and or beverages in the laboratory is simply a matter of safety and regulated by both state law. The laboratory is never a place to eat or drink anything; one mistake can cause serious permanent harm or kill you. Laboratories are filled with poisonous materials, don’t risk ingesting them.

Late Assignments and Laboratory Reports:
Laboratory reports must be handed in on time. A penalty of one point per day will be taken for late reports. No laboratory reports or any other exercise will be graded for credit if received after 5 PM, on Thursday, May 05, 2022.

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List of Reading Assignments:

RA02: C. P. Opeil and M. J. Graf, "Evidence for the existence of a magnetic quantum critical point in $U(Pt_{1-x}Pd_x)_3$" *Physica B* **319**, 246-250 (2002).

RA03: C. P. Opeil, A. de Visser, M. J. Naughton, and M. J. Graf "Crossover from anomalous to conventional antiferromagnetism in Pd-doped UPt_3 studied via cantilever magnetometry" *Journal of Magnetism and Magnetic Materials* **272-276**, 244-245 (2004).

RA04: C. P. Opeil, J. C. Lashley, J. L. Smith "Angle-dependent magnetoresistance near the premartensitic phase of Ni_2MnGa" *Journal of Physics: Conference Series* **150** 042109 (2009).

RA05: C. P. Opeil, B. Mihaila, R. K. Schulze, L. Manosa, A. Planes, W. L. Hults, R. A. Fisher, P. S. Riseborough P. B. Littlewood, J. L. Smith, and J. C. Lashley "Combined experimental and theoretical investigation of the premartensitic transition in Ni_2MnGa" *Physical Review Letters* **100**, 165703 (2008).

RA06: C. P. Opeil, R. C. Albers, K. B. Blagoev, M. Gulácsi, J. C. Lashley, P. B. Littlewood, M. E. Manley, B. Mihaila, P. S. Riseborough, R. K. Schulze, D. J. Thoma, H. M. Volz and J. L. Smith "Photoelectric effect in Uranium" *Journal of the Physical Society of Japan* Vol. **75**, 56-57 (2006).

RA07: C. P. Opeil, R. K. Schulze, H. M. Volz, J. C. Lashley, M. E. Manley, W. L. Hults, R. Hanrahan, Jr., J. L. Smith, B. Mihaila, K. B. Blagoev, R. C. Albers, P. B. Littlewood "Angle-resolved photoemission and first-principles electronic structure of α-U(001)" *Physical Review B*, **75**, 045120 (2007).

RA08: C. P. Opeil, G.J. Consolmagno, D.T. Britt "The thermal conductivity of meteorites: New measurements and analysis" *Icarus* **208**, 449-454 (2010).

RA09: K. C. Lukas, W.S. Liu, Q. Jie, Z.F. Ren, C.P. Opeil, "Thermal Stability of Thermoelectric Materials via In Situ Resistivity Measurements" *Review of Scientific Instruments* **83**, 115114, 2012.

RA10: K. C. Lukas, W. S. Liu, G. Joshi, Z. F. Ren, C. P. Opeil "Thermoelectric Properties of Ho Doped $Bi_{0.88}Sb_{0.12}$" *Journal of Materials Science*, **47**, 5729-5734 (2012).

RA11: K. C. Lukas, W. S. Liu, Z. F. Ren, and C. P. Opeil "Transport properties of Ni, Co, Fe, Mn doped $Cu_{0.01}Bi_2Te_{2.7}Se_{0.3}$ for thermoelectric device applications" *Journal of Applied Physics* **112**, 054509, 2012.

RA12: H. Zhao, M. Pokharel, G. Zhu, S. Chen, K. Lukas, Q. Jie, C. Opeil, G. Chen, and Z. Ren "Dramatic thermal conductivity reduction by nanostructures for large increase in thermoelectric figure-of-merit of $FeSb_2$" *Applied Physics Letters*, **99**, 163101 (2011).

RA13: C. P. Opeil, D. T. Britt, R. J. Macke and G. J. Consolmagno, "The Surprising Thermal Properties of CM Carbonaceous Chondrites" *Meteoritics & Planetary Science* **55**, E1–E20 (2020).

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Academic Calendar Boston College Spring 2022

January 18: (Tuesday) UG Classes begin
March 07 (Monday) - March 11 (Friday): Spring Break
April 14: (Thursday) Holy Thursday no classes after 4 PM
April 15: (Friday) Good Friday no classes
April 18: (Monday) Patriot's Day, no classes
April 19: (Tuesday) Substitute Monday class schedule
May 06 (Friday) & May 09 (Monday):
Study days - No classes for undergraduate students
May 11 (Wednesday) - May 17 (Tuesday): Term examinations
May 23: (Monday) BC Commencement

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2022 CALENDAR

www.wiki-calendar.com

JANUARY						
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FEBRUARY						
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APRIL						
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MAY						
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JUNE						
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AUGUST						
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OCTOBER						
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NOVEMBER						
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DECEMBER						
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HOLIDAYS AND OBSERVANCES:

Jan 1 - New Year's Day
Jan 17 - Martin Luther King Jr. Day
Feb 14 - Valentine's Day
Feb 21 - Presidents' Day
Mar 17 - St. Patrick's Day
Apr 17 - Easter Sunday
Apr 18 - Easter Monday
Apr 18 - Tax Day
May 5 - Cinco de Mayo
May 8 - Mother's Day
May 30 - Memorial Day
Jun 14 - Flag Day
Jun 19 - Father's Day
Jun 19 - Juneteenth
Jun 20 - 'Juneteenth' day off
Jul 4 - Independence Day
Sep 5 - Labor Day
Oct 10 - Columbus Day
Oct 31 - Halloween
Nov 8 - Election Day
Nov 11 - Veterans Day
Nov 24 - Thanksgiving Day
Nov 25 - Black Friday
Dec 24 - Christmas Eve
Dec 25 - Christmas Day
Dec 26 - 'Christmas Day' day off
Dec 31 - New Year's Eve

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Semester Schedule: Spring 2021
Tuesday & Thursday 1:30-4:00 PM
Office Hours: Due to the COVID19 pandemic all office hours will be via Zoom. Please, email the Instructor at opeil@bc.edu to set up a Zoom meeting for Tuesday 4:00-5:30 PM or Thursday 4:00-5:30 PM. Offices hours are only offered during these times.

Experimental Laboratory Procedure - Abbreviations:
1. Using an ULVAC-ZEM to measure Seebeck and resistivity on Constantan and Nickel. [ZSR]
2. Speed of Sound in Metals at 295 K and Young's Modulus, multi-frequency probe. [SSMF]
3. Strain gage calibration and measurement. [SGCM]
4. Speed of Sound in materials at 295 K using a single frequency, through transmission. [SSTT]
5. Capacitance and Dielectric Measurements & Measuring Q in a Tank circuit. [CQTT]
6. Measuring resistivity of a metal at T = 300 to 4 K with B = 0 and 90,000 Oe. [RMTB]
7. Measuring diamagnetism or paramagnetism in elements. [MDPE]

Date	Lecture or Laboratory	Detail	Homework Assignment	Laboratory Report	Reading Quiz
Tuesday, January 18, 2022	Lecture 1, 1:30-4:00 PM Higgins 250	Classes begin: Syllabus review, Laboratory Safety review.	Library Exercise assigned	Laboratory reports are due one week after the end of that laboratory experiment.	N/A
Thursday, January 20, 2022	Lecture 2, 1:30-4:00 PM Higgins 250	Pre-labs for experiments and theory.	Exercise part 1 Archimedes Principle assigned Library Exercise due at 5:00 PM		N/A
Tuesday, January 25, 2022	Laboratory 1, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.	Kaleidagraph Exercise assigned		
Thursday, January 27, 2022	Laboratory 1, 1:30-4:00 PM Higgins 130 part 2		Exercise part 1 Archimedes Principle due at 5:00 PM		RA02

Tuesday, February 01, 2022	Laboratory 1, 1:30-4:00 PM Higgins 130 part 3				
Thursday, February 03, 2022	Laboratory 1, 1:30-4:00 PM Higgins 130 part 4		Kaleidagraph Exercise due at 5:00 PM		RA03
Tuesday, February 08, 2022	Laboratory 2, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.		Laboratory reports due.	
Thursday, February 10, 2022	Laboratory 2, 1:30-4:00 PM Higgins 130 part 2				RA04
Tuesday, February 15, 2022	Laboratory 2, 1:30-4:00 PM Higgins 130 part 3				
Thursday, February 17, 2022	Laboratory 3, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.	C_p Exercise assigned.		RA05
Tuesday, February 22, 2022	Laboratory 3, 1:30-4:00 PM Higgins 130 part 2			Laboratory reports due.	
Thursday, February 24, 2022	Laboratory 3, 1:30-4:00 PM Higgins 130 part 3		C_p Exercise due at 5:00 PM		RA06
Tuesday, March 01, 2022	Laboratory 4, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.			

Thursday, March 03, 2022	Laboratory 4, 1:30-4:00 PM Higgins 130 part 2			Laboratory reports due.	RA07
Tuesday, March 15, 2022	Laboratory 4, 1:30-4:00 PM Higgins 130 part 3				
Thursday, March 17, 2022	Laboratory 5, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.			RA08
Tuesday, March 22, 2022	Laboratory 5, 1:30-4:00 PM Higgins 130 part 2			Laboratory reports due.	
Thursday, March 24, 2022	Laboratory 5, 1:30-4:00 PM Higgins 130 part 3				RA09
Tuesday, March 29, 2022	Laboratory 6, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.			
Thursday, March 31, 2022	Laboratory 6, 1:30-4:00 PM Higgins 130 part 2			Laboratory reports due.	RA10
Tuesday, April 05, 2022	Laboratory 6, 1:30-4:00 PM Higgins 130 part 3				
Thursday, April 07, 2022	Laboratory 7, 1:30-4:00 PM Higgins 130 part 1	Pre-lab for experiments and theory.			RA11

Tuesday, April 12, 2022	Laboratory 7, 1:30-4:00 PM Higgins 130 part 2			Laboratory reports due.	
Thursday, April 21, 2022	Laboratory 7, 1:30-4:00 PM Higgins 130 part 3				RA12
Tuesday, April 26, 2022	Student presentation				
Thursday, April 28, 2022	Make up lab days or Archimedes 2				RA13
Tuesday, May 03, 2022	Make up lab days or Archimedes 2			Laboratory reports due.	
Thursday, May 05, 2022	Make up lab days or Archimedes 2			All laboratory reports due by 5 PM	No reading quiz